



Pairing geophysical techniques improves understanding of the near-surface Critical Zone: Visualization of preferential routing of stemflow along coarse roots

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ABSTRACT

There is compelling evidence from aboveground observations that trees considerably affect precipitation partitioning through the Critical Zone. However, due to the lack of appropriate methods, the role of root systems (the hidden half of trees) on redistributing precipitation and infiltration into and routing through the soil remains inadequately visualized and understood. Here, we designed a novel experiment to pair two non-invasive geophysical techniques, ground-penetrating radar (GPR) and electrical resistivity tomography (ERT), to trace stemflow through the soil in a forested hillslope after water was released on the trunk of an American beech (*Fagus grandifolia* Ehrh.) to induce stemflow. We used GPR to locate lateral coarse roots and GPR and ERT together to confirm the wetting areas in response to the non-uniform transport of stemflow. Signal changes between time-lapse geophysical images were used to reveal both the response time and location of stemflow infiltration into and redistribution through the soil. This first known study to investigate the subsurface routing of stemflow by combining GPR and ERT revealed that the belowground funneling of stemflow along laterally oriented coarse roots transported water 2.8 m downslope from the study tree in 30 min after stemflow was initiated. In situ excavation validated the distribution of lateral roots and lateral root-derived preferential flow paths identified in geophysical images, confirming the utility of pairing GPR and ERT to gain insights into the temporal dynamics and spatial distribution of subsurface routing of stemflow. The proposed method visualized and confirmed the funneling effect of roots on belowground water redistribution that contributed to subsurface lateral flow. Pairing GPR and ERT provides a useful combination of geophysical methods to advance our understanding of the complex interactions between plant and soil, such as the role of tree roots in soil hydrological process by revealing areas of funneling in the hidden part of the Critical Zone.

1. Introduction

Biodiversity and ecosystem services are strongly tied to water flux through the Critical Zone, the thin layer of the Earth's terrestrial surface that ranges from the top of the vegetation canopy to the soil and down to the weathering zone (Guo and Lin, 2016). Trees play an array of roles in storing, partitioning, and redistributing precipitation, thereby influencing the hydrologic cycle as well as the associated pedological and

biogeochemical processes in the Critical Zone (Levia and Germer, 2015; Spencer and van Meerveld, 2016). Johnson and Lehmann (2006) introduced a double-funneling model (Fig. 1) to describe the redistribution of precipitation by trees above the ground and directing non-uniform routing in the soil. Better understanding the double-funneling process helps explain the heterogeneous nature of the chemical, biological, and hydrological properties of the soil (Gersper and Holwaychuk, 1971; Germer, 2013; Li et al., 2009) and model the

Abbreviations: ERT, electrical resistivity tomography; GPR, ground-penetrating radar; LAI, leaf area index; MD, Maryland; RHT, Randomized Hough Transform; SS, structural similarity; TL, time-lapse

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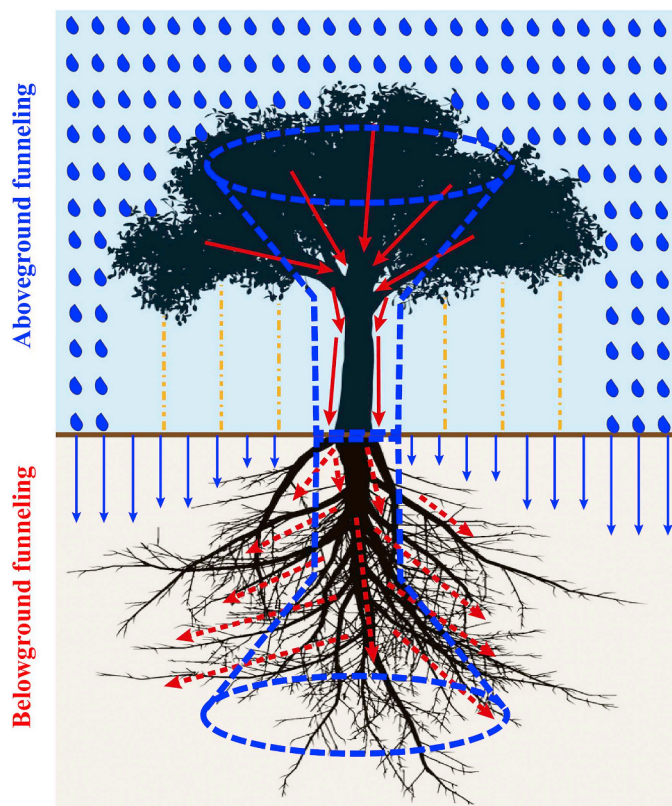


Fig. 1. The double-funneling model of trees redistributing precipitation both above and belowground. Red arrows indicate the incident precipitation that is concentrated by tree stems as stemflow (i.e. aboveground funneling). Yellow dashed lines indicate the throughfall. Red dashed arrows indicate the non-uniform routing of precipitation along tree roots as preferential flow (i.e. belowground funneling), which has a faster flow velocity and greater percolation depth than the uniform infiltration and percolation into the soil (indicated by the blue arrows). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

complex plant-soil relationship as well as the interaction between the aboveground and the belowground parts of the Critical Zone (Carlyle-Moses et al., 2018; Guo and Lin, 2016).

Many studies have investigated the aboveground funneling effect whereby the incident precipitation is spatially concentrated by tree stems via stemflow (Fig. 1; e.g. Carlyle-Moses and Price, 2006; Herwitz, 1986; Levia and Germer, 2015). Localized concentrations of stemflow input near tree trunks produce persistent patterns of wet and dry areas on the forest floor (Keim et al., 2005) or in the soil (Li et al., 2009), change soil pH and induce soil acidification (Falkengrengrerup, 1989), enhance soil microbial activity (Rosier et al., 2016), and lead to the spatial variation of radiocesium stocks in the soil (Imamura et al., 2017) as well as the development of perched water tables in the shallow soil (Germer, 2013).

Belowground, stemflow inputs funneled along roots create non-uniform flow (Fig. 1; Levia and Frost, 2003), and influence the soil water budget at the pedon scale as well as stormwater runoff, groundwater recharge, and stream generation at the hillslope and catchment scales (Beven and Germann, 2013; Germer, 2013; Lin, 2010). However, due to the lack of appropriate methods and the heterogeneity and hidden structures in the soil, our understanding of the belowground funneling of precipitation and the influence of roots in subsurface water transport remains insufficient (Guo and Lin, 2018). Prior work used dye tracers to examine preferential channelization of stemflow along roots, which recognized the critical role of belowground funneling of stemflow in soil moisture regimes (e.g. Li et al., 2009; Liang et al., 2011; Schwärzel et al., 2012; Spencer and van Meerveld,

2016). However, the dye staining patterns only provide one-time results after the event and preclude a temporal account of the development of flow dynamics (Guo and Lin, 2018). The sorption feature of dye tracers to soil constituents also limits their ability to represent actual water flow pathways (Allaire et al., 2009). Therefore, new approaches of revealing the ecohydrological process in the soil, such as water transport along roots, are needed to test the double-funneling model, especially the belowground funneling.

Geophysical techniques have emerged as valuable means for non-invasive detection of coarse roots (> 5 mm in diameter) and subsurface flow pathways (Jayawickreme et al., 2014; Parsekian et al., 2015). Among the common geophysical methods, ground-penetrating radar (GPR) that detects the contrast in dielectric permittivity in the subsurface offers the best combination of detection depth, spatial coverage, and time-space resolution (Parsekian et al., 2015). Since water content exerts a first-order influence on dielectric permittivity, GPR wave velocity and reflection strength are sensitive to water variation in the subsurface (Guo et al., 2013). The higher water content in roots relative to the surrounding soil matrix ensures the necessary contrast in dielectric permittivity, which makes root detection by GPR possible (Butnor et al., 2003). So far GPR has been proven effective in identifying root branching patterns (Guo et al., 2015; Zhu et al., 2014), reconstructing root system architecture and root profiles (Borden et al., 2017; Wu et al., 2014; Cui et al., 2019), and measuring root biomass (Butnor et al., 2003; Guo et al., 2013; Liu et al., 2018; Molon et al., 2017) and root zone soil moisture (Liu et al., 2019). However, previous GPR-based root investigation was primarily based on a one-time survey and could not reveal the occurrence and dynamics of the belowground funneling process.

Recently, time-lapse GPR has been adopted to map subsurface flow pathways (Angermann et al., 2017; Guo et al., 2014; Jackisch et al., 2017; Guo et al., 2019) via identifying areas with signal differences between subsequent GPR images (Holden, 2004; Doolittle et al., 2012). Given the potential of GPR in detecting coarse roots, time-lapse GPR is promising to monitor belowground funneling along roots. However, repeating GPR surveys with identical geometry is tricky in the field. The geometric mismatch between time-lapse images leads to signal changes in the differenced GPR images (i.e. subtracting the background image from post-wetting images) and impairs the accuracy of mapping flow pathways (Guo et al., 2014). Besides, it remains challenging to convert GPR reflection strength to soil moisture (Huisman et al., 2003), which makes time-lapse GPR difficult to quantitatively represent soil moisture changes. Ancillary measurement is, thus, required to assist reasonable interpretations of time-lapse GPR. For example, Guo et al. (2014) used an array of sensors to monitor soil moisture dynamics to constrain GPR results. To ensure the quality of GPR detection, soil sensors were installed in a trench downslope of the GPR survey area so that the direct evidence of changes in soil moisture in the survey area was still lacking. Therefore, in addition to traditional instrumentation and ground-truthing, such as borehole data or extensive excavation, which is destructive and labor-intensive, complementary geophysical methods should be integrated to enhance reliable and non-invasive characterization of belowground funneling as well as other subsurface ecohydrological processes (Leucci and De Giorgi, 2005).

Another commonly used geophysical technique is electrical resistivity tomography (ERT), which generates low-frequency electrical currents to measure electrical resistivity distributions in the subsurface. The electrical resistivity of soil is strongly dependent on soil porosity, pore-water saturation, the conductivity of the pore water, and soil texture and mineralogy (Parsekian et al., 2015). Thus, ERT is sensitive to changes in water distribution in the subsurface (Michot et al., 2003). To date ERT has been used successfully in mapping the root zone (Davidson et al., 2011; Leucci, 2010), quantifying root biomass (Amato et al., 2008), locating preferential flow influencing area (Travalletti et al., 2012), and monitoring soil moisture dynamics in the root zone (Robinson et al., 2012). As the electrodes for data collection remain in the same location throughout the survey, time-lapse ERT images from

the same survey line have identical geometry. The signal changes between time-lapse ERT images represent the variation of electrical resistivity. Due to the close relationship between soil moisture and electrical resistivity, time-lapse ERT captures the trend of soil moisture change over time. If such a relationship is calibrated based on site-specific conditions, time-lapse ERT can quantitatively depict the temporal evolution of soil moisture (Michot et al., 2003).

However, the coarser detection resolution of ERT with respect to GPR makes it very challenging to separate individual roots or flow pathways in ERT images. Also, repeating an ERT scan usually takes longer time than repeating a GPR scan for the same survey length. Therefore, GPR (with a fine resolution to locate individual roots and a fast data acquisition rate to capture transient dynamics in the subsurface) and ERT (revealing the spatial pattern of subsurface moisture change) should be combined to facilitate non-invasive monitoring of preferential flow along coarse roots. If the areas with signal changes in time-lapse GPR images relative to pre-wetting condition also show changes in electrical resistivity, it is very likely that these areas are locations of flow pathways with strong moisture fluctuation.

We thus hypothesize that through a refined experimental design of pairing multiple approaches, improved time-lapse geophysical surveys can reveal the areas of funneling of stemflow in the subsurface. To test this hypothesis, we designed a novel experiment to combine GPR and ERT to trace stemflow through the soil in a forested hillslope. This study aims to demonstrate the utility of combining time-lapse GPR and ERT to trace the non-uniform routing of stemflow along lateral coarse roots over time and through space to gain a more thorough process-based understanding of stemflow entering the soil.

2. Materials and methods

2.1. Study site description and the sample tree

The study site is located within the Fair Hill Natural Resources Management Area (39°43' N, 75°50'W) in northeastern Maryland, USA (Fig. 2a, b). This area has a humid, continental climate, with a mean annual air temperature of 13 °C and mean annual precipitation around 1200 mm (Levia et al., 2010). A total of 2.4 mm of precipitation was recorded in the week preceding the experiment. Fair Hill is underlain by the Mt. Cuba Wissahickon formation, with the dominant soil being mapped as the Glenelg series (*coarse-loamy, mixed, mesic Lithic Dystrudepts*) (Inamdar et al., 2013). The forest canopy is deciduous with *Fagus grandifolia* Ehrh. (American beech), *Liriodendron tulipifera* L. (yellow poplar), and *Acer rubrum* L. (red maple) among the dominant species. The stand density and summer leaf area index (LAI) is 225 trees ha⁻¹ and 5.3 m² m⁻², respectively (Siebert and Levia, 2014).

An American beech (with the diameter at breast height of 74.9 cm and a basal area of 4406 cm²) (Fig. 2e), which was previously instrumented for stemflow measurement (Levia et al., 2010), was selected for the experiment. The study tree is situated on a north-facing hillslope with a gentle slope gradient of 12° and an approximate elevation of 80 m a.s.l. (Fig. 2c). Our geophysical survey covered the downslope area of the sample tree (Fig. 2c, f). The upslope section of the survey area (i.e. near the sample tree) is underlain by well-drained, coarse loam to fine sandy loam (Fig. 2d), where the soil has an 8 cm thick A horizon and a weakly-developed Bw horizon. Most lateral roots are distributed between 10 and 50 cm depth, with the boundary to the C horizon being approximately 70 cm depth. Whereas the downslope section of the survey area (i.e. downslope away from the sample tree) is underlain by poorly-drained, fine loam with redox features, having a deeper boundary to the C horizon (~1 m depth) that allows a greater rooting depth of lateral roots. The average percentage of sand, silt, and clay of the soil in the survey area is 25 ± 7.5%, 53 ± 4.3%, and 22 ± 5.0%, respectively. At the base of the hillslope (6 m downslope of the tree), a second-order stream flows into the Big Elk Creek and eventually to the Chesapeake Bay (Fig. 2a, b).

2.2. Stemflow routing and time-lapse geophysical investigations

The stemflow routing experiment and time-lapse geophysical investigations were carried out on September 18, 2017, about two weeks after Hurricane Harvey brought heavy rainstorms to the study region. A survey grid (6.0 m × 2.8 m) consisting of three GPR survey lines and two ERT survey lines was established perpendicular to the hillslope (Fig. 2c). The base of the study tree extended from horizontal locations of 3.0 to 4.0 m along the survey line. Both ends of the GPR survey line were anchored into the ground to ensure the reproducibility of time-lapse data. Prior to the experiment, two GPR and ERT surveys were conducted to record the initial soil condition (Table 1).

The stemflow input scheme was designed to not only deliver a realistic total volume of stemflow based on the relationship between open-land precipitation and stemflow measured for the sample tree in 2007–2008 (Levia et al., 2010) but to also ensure the development of a wetting front that was detectable by both GPR and ERT. Approximately 100 L (corresponding to 12 mm of incident precipitation) was applied to the trunk over a period of 55 min in five pulses of water release (each time with 20 L of water) that were carried out every ten to fifteen minutes to allow for geophysical measurement at set intervals to follow the development of a wetting front and its downslope progress (Table 1). The forest litter around the base of the sample tree was not removed to enable a better representation of stemflow entering into the soil under natural conditions. During the experiment, regular tap water was released directly on the tree trunk at the breast height facing downslope through a PVC tubing to create stemflow (Fig. 2e). The PVC tubing was connected to a water tank, and the flow rate was approximately 4 L min⁻¹. Overland flow was virtually non-existent (extending < 0.5 m downslope), with transient localized ponding at the base of the tree, where most of the stemflow infiltrated. All surface water had infiltrated before entering the geophysical survey area. One GPR survey and one ERT survey were performed immediately after each water release and repeated four more times after the last water release (Table 1), leading to a total of eleven time-lapse surveys. Near real-time measurements during an actual stemflow event are impractical, and thus the timing and volume of water delivered were controlled to allow for meaningful analysis and process-based insights of the double funneling effect.

A 750 MHz Mala GX GPR system (ABEM Mala Geoscience, Mala, Sweden) in the common offset mode (0.16 m antenna separation) was used to collect data at 0.02 m intervals along each survey line. The center frequency of a GPR system determines the detection resolution and penetration depth. A higher frequency picks up finer structural details in the subsurface but has shallower penetration, whereas a lower antenna frequency allows a deeper penetration depth but a coarser spatial resolution (Guo et al., 2013). The 750 MHz antennas were the optimal combination of detection resolution (i.e. image quality), depth penetration, and the convenience of operation in the sloping area based on testing antenna frequencies from 250 to 2300 MHz under the study site conditions. The selected GPR system had a high signal-to-noise ratio above 0.8 m depth. An IRIS Syscal Kid system (IRIS Instruments, Orléans, France) was programmed to collect alternating Wenner array measurements of the ERT data over two survey lines (each with 12 electrodes arranged at a 0.5 m spacing). The Wenner array was chosen due to its broad ability to resolve both lateral and vertical changes in resistivity to monitor soil moisture dynamics and identify preferential flow influencing zones, which can have a vertical resolution at half the distance of the electrode spacing, i.e. 0.25 m (Binley and Kemna, 2005). Although a narrow spacing between electrodes can improve the detection resolution of ERT, it also takes longer time for data collection and cannot match the survey interval of repeating time-lapse GPR.

2.3. Geophysical data post-processing

To enhance the clarity of GPR images, the following procedures were applied per Guo et al. (2014): (1) Removal of data collected at the

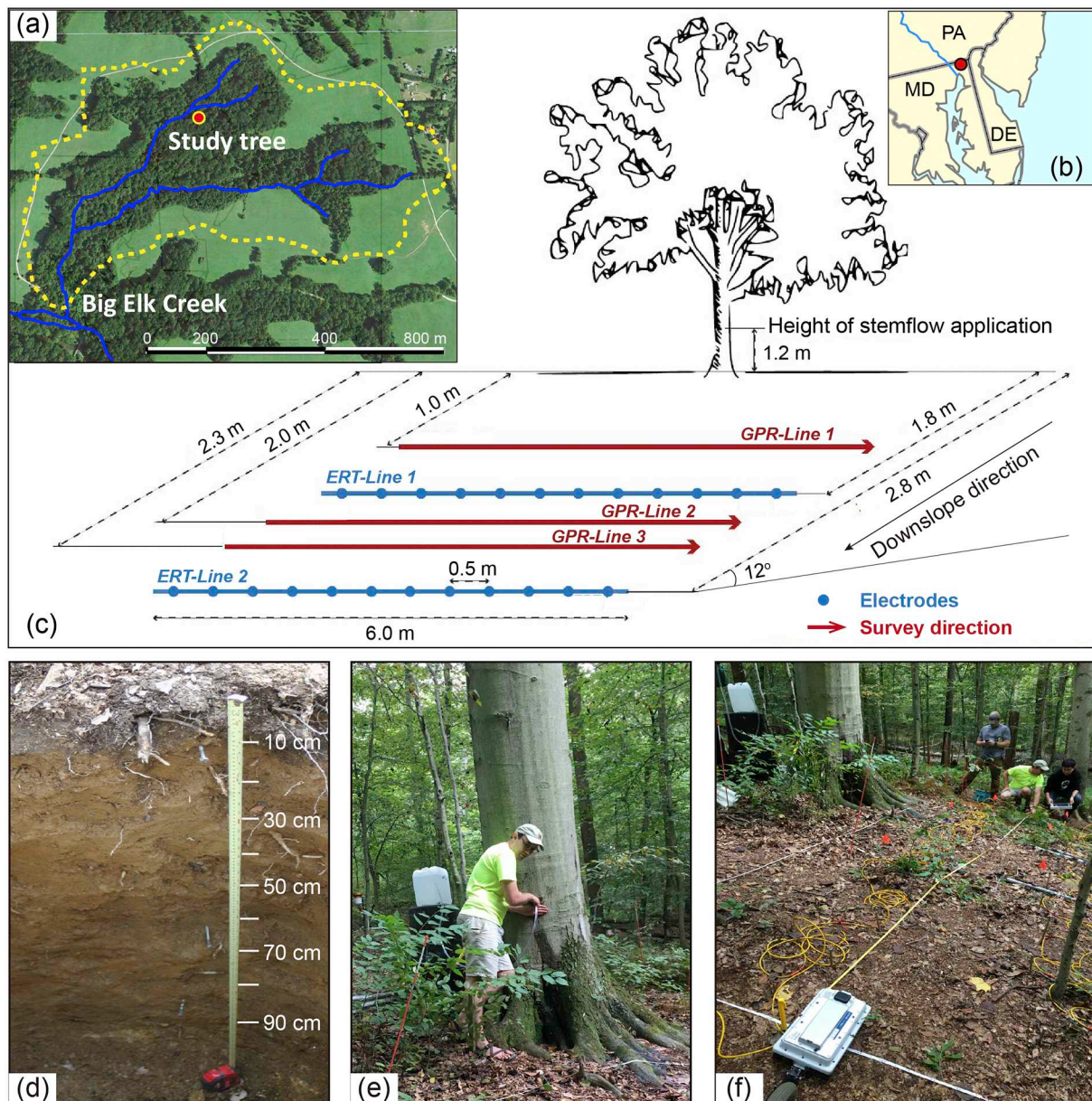


Fig. 2. (a) An overview of the Fair Hill Catchment, showing drainage boundaries (indicated by the yellow dashed lines) and the location of the sample tree. (b) This catchment is located in the Piedmont physiographic region of Maryland (MD) and drains into the Chesapeake Bay. (c) The schematic of the geophysical survey grid. The base of the study tree extends 3.0 to 4.0 m along the survey line. (d) A photo of the typical soil type (the Glenelg series) in the catchment, showing a 10 cm organic layer and abundant lateral roots from the surface to 50 cm in depth. (e) Water was released on the tree trunk through a PVC tubing to represent the stemflow process. (f) A photo of the geophysical survey area, showing the antenna of the GPR system (the white box) and the cables connecting the electrodes of the ERT system. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

edges (0–0.1 m and 5.5–6.0 m) of the survey line to eliminate the influence of a soil trench on the quality of the GPR data, which was excavated 0.5 m away from the end of the survey line to constrain the interpretations of the geophysical results; (2) First-arrival time zero adjustment to suppress the effect of uneven topography on data reproducibility; (3) Removal of data collected after 20 ns (two-way travel time) due to a low signal-to-noise ratio; (4) Travel time to depth conversion by a wave velocity of 8.0 cm/ns, which was estimated using a dielectric constant (15.18) derived from the Topp's equation based on initial soil moisture reading. Thus, the penetration depth of the used GPR system at the study site was 0.8 m. Because of the heterogeneity in the soil and the spatial variation of the wetting pattern after stemflow infiltration, wave velocity was simplified to be constant across soil depths (Nyquist et al., 2018; Truss et al., 2007); (5) Time-based

exponential scaling to compensate energy attenuation with depth; and (6) First-order Butterworth filter and Dewow filter to suppress the level of signal noise. All steps were performed with Reflex-Win 7.2 (Sandmeier Scientific Software, Karlsruhe, Germany).

The Randomized Hough Transform (RHT) algorithm was used for lateral coarse root identification by detecting hyperbolic reflections in GPR images (Li et al., 2016). Automatically identified root signals by the RHT algorithm were visually checked to ensure detection accuracy. The peak of a hyperbolic reflection in GPR images indicates the location of a lateral coarse root (Guo et al., 2015). Only the hyperbolae that constantly appeared throughout all time-lapse surveys were retained for identifying and locating coarse roots.

For preferential flow detection, we standardized the repeated GPR images collected over the same survey line into an identical geometry

Table 1

Timing and sequences of events during the time-lapse geophysical surveys and the stemflow routing experiment.

Time	Time-lapse surveys	Elapsed time (min)	Event
11:40–11:45	First (TL1) ^a	0	Pre-wetting background GPR and ERT surveys
11:50–11:55	Second (TL2)	10	Repeat of pre-wetting background surveys
12:00–12:05		20	First release of 20 L of tap water
12:10–12:15	Third (TL3)	30	GPR and ERT surveys after the first water release
12:15–12:20		35	Second release of 20 L of tap water
12:25–12:30	Fourth (TL4)	45	GPR and ERT surveys after the second water release
12:30–12:35		50	Third release of 20 L of tap water
12:35–12:40	Fifth (TL5)	55	GPR and ERT surveys after the third water release
12:40–12:45		60	Fourth release of 20 L of tap water
12:45–12:50	Sixth (TL6)	65	GPR and ERT surveys after the fourth water release
12:50–12:55		70	Fifth release of 20 L of tap water
12:55–13:00	Seventh (TL7)	75	GPR and ERT surveys right after the fifth water release
13:35–13:40	Eighth (TL8)	115	GPR and ERT surveys 40 min after the fifth water release
14:10–14:15	Ninth (TL9)	150	GPR and ERT surveys 75 min after the fifth water release
14:40–14:45	Tenth (TL10)	180	GPR and ERT surveys 105 min after the fifth water release
15:00–15:05	Eleventh (TL11)	200	GPR and ERT surveys 125 min after the fifth water release

^a TL stands for time-lapse.

(Guo et al., 2014), and calculated the structural similarity (SS) between pre-wetting background and post-wetting images to identify areas where soil moisture has increased (Allroggen and Tronicke, 2016; Wang et al., 2004):

$$SS(x, y) = \frac{\sigma_{(Ax, Ay)} + a}{\sigma_{Ax} \sigma_{Ay} + a}, \quad (1)$$

$$\sigma_{(Ax, Ay)} = \frac{1}{N-1} \sum_{i=1}^N (Ax_i - \mu_{Ax})(Ay_i - \mu_{Ay}), \quad (2)$$

where x and y are two data acquisition times, σ_{Ax} and σ_{Ay} are standard deviations of amplitudes (A) of GPR signal in time sequences Ax and Ay , μ_{Ax} and μ_{Ay} are mean amplitudes, and $\sigma_{(Ax, Ay)}$ indicates the covariance between Ax and Ay . To avoid numerical instabilities due to near-zero σ values, a with the value of 10% of the maximum amplitude is introduced. As suggested by Allroggen and Tronicke (2016), we calculated SS using a local Gaussian window of 2.5 ns along the vertical axis and 0.1 m along the horizontal axis. N is the total number of samples in the windowed sequence, and i is the rank of the sample. The calculated SS ranges from -1 to 1 , with 1 indicating no change between two GPR images and -1 meaning the most dissimilar (Angermann et al., 2017; Jackisch et al., 2017). If the wetting areas overlapped the detected roots, we considered this as evidence of preferential transport of stemflow along lateral roots.

The ERT apparent conductivity data was downloaded from the instrument and reviewed in the IRIS Prosys II software package (IRIS Instruments, Orleans, France) for general trends and errors prior to inversion. For consistency, we reviewed the data for errors due to bad electrode contact with the soil and confirmed the electrode spacing and sequence values were correct. Secondly, we checked the data points that were $> 2.0\%$ of the stacking standard deviation for each apparent resistivity value reported. However, none of the measurements approached that threshold and were removed. All ERT apparent conductivity data was inverted to resistivity values in the Res2dinv software package (Geotomosoft Solutions, Penang, Malaysia), which uses a finite-difference routine to calculate apparent resistivity and then a non-linear smoothness-constrained least-squares regression to calculate resistivity (Degroot-Hedlin and Constable, 1990). While it was impossible to collect reciprocal measurements due to the nature of the experiment design to determine the fast flow process, root mean square error of the inverted resistivity data was approximately 2.5%. Due to the relatively flat topography over the geophysical survey area, we did not do terrain correction for both GPR and ERT images.

2.4. Ground truthing of interpretations of geophysical results

In order to identify subsurface features that may influence geophysical signals and constrain the interpretations of the geophysical data, a soil trench (0.3 m wide, 0.6 m deep, and 4.5 m long) was opened along the survey slope (0.5 m away from the geophysical scanning plot) to examine the soil profile and rooting depth and observe the arrival of subsurface flow during the experiment. After the completion of the experiment, four additional test locations were excavated. The first was located between the base of the sample tree to GPR-Line 2, which was carefully excavated to the depth of 20 cm at 2 cm increments to expose and document the distribution of roots and identify GPR reflector locations. The soil beneath GPR-Line 2 was examined similarly in three additional soil pits that reached 40 cm depth (the bottom of the primary rooting zone). The results of the post-survey excavations were used as a direct comparative dataset to aid in the interpretations of the geophysical results and to collaborate any reflectors to heterogeneous areas or isolated reflector.

3. Results

3.1. Time-lapse GPR

Time-lapse GPR images collected before any water release, immediately after the last water release, and under the final condition are compared in Fig. 3. For the soil below GPR-Line 1, reflection strength shows a significant increase at 0.5 to 3.0 m along the survey line (indicated by the blue dashed box in Fig. 3a–c), where the most lateral roots are distributed (the yellow dots in Fig. 3d). Structural similarity (SS) between the pre-wetting and post-wetting images depicts the areas with increased soil moisture (the blue areas in Fig. 3e–f). The wetting areas are mostly within the root zone, affirming that the root zone soil is more receptive to subsurface flow. Some isolated areas with strong soil moisture increases (the dark blue patches) overlap with the detected lateral roots (the blue dots in Fig. 3e–f), suggesting that after entry into the soil, a portion of the stemflow was funneled along the lateral roots. In total, 10 out of 27 (37%) identified lateral roots contributed to the downslope transport of stemflow under the final condition of the experiment (Fig. 3f).

As the distance from the base of the tree increases downslope, areas showing higher signal strength become more limited (the blue dashed box in Fig. 3g–i, m–o). This suggests a reduced amount of stemflow reached the downslope section of the survey area. There are also isolated wetting areas identified on the SS map (the blue areas in Fig. 3k–l, q–r) overlapping with the detected lateral roots (the blue dots in Fig. 3k–l, q–r). However, lateral roots that directed stemflow water

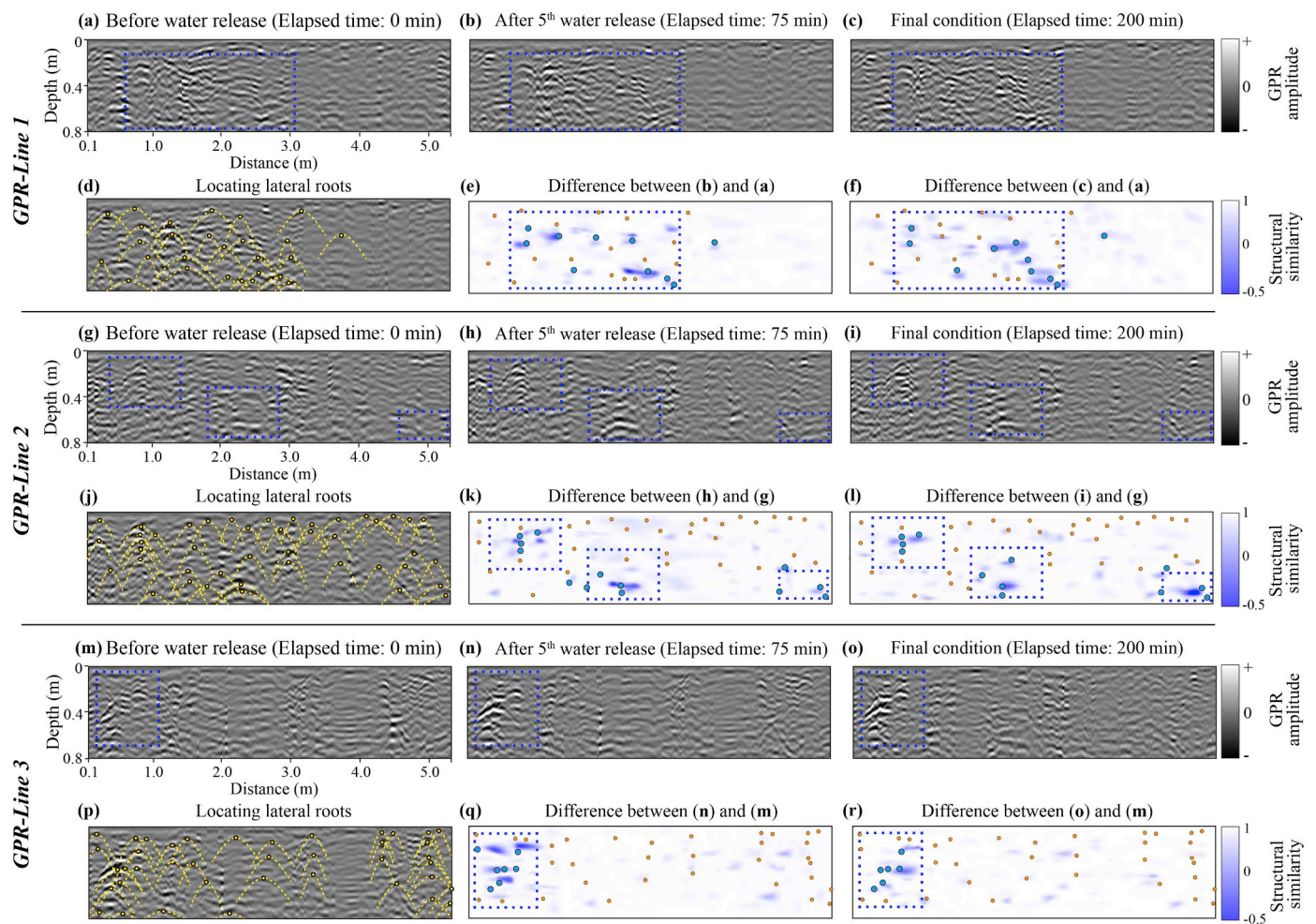


Fig. 3. Time-lapse GPR images collected (1) before stemflow application over GPR-Line 1 (a), GPR-Line 2 (g), and GPR-Line 3 (m), (2) immediately after the fifth water release (b, h, and n), and (3) under the final condition (c, i, and o). (d), (j), and (p) show lateral roots detected in GPR images. The peak of the hyperbola indicates the location of a lateral root. (e), (k), and (q) show the similarity attributes (SS) calculated between the pre-wetting GPR image and that after the fifth water release. (f), (l), and (r) show the SS between the pre-wetting and final conditions. Lower SS values indicate the higher deviation from the pre-wetting condition, which is likely caused by the increase in soil moisture due to subsurface preferential flow. The blue dashed boxes highlight the areas showing greater changes in reflection patterns in time-lapse GPR images. The blue dots in (e, f, k, l, q, and r) refer to the lateral roots that are overlapping with the isolated wetting areas (i.e. the identified preferential flow pathways), evidencing the lateral root-derived preferential flow. The orange dots in (e, f, k, l, q, and r) indicate the lateral roots that do not contribute to preferential channelization of stemflow. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

beneath GPR-Line 2 and Line 3 are fewer than those below Line 1. Only 13 out of 44 (29%) or 7 out of 40 (17%) lateral roots contributed to stemflow transport below GPR-Line 2 and Line 3 under the final condition, respectively.

The temporal evolution of the preferential transport of stemflow in the subsurface is compared at each of the time-lapse GPR surveys (Figs. S1–S3). High SS values between two background surveys (Fig. 4; also see TL2–TL1 in Figs. S1–S3) demonstrate the high reproducibility of the repeated GPR images. Right after the second water release, areas with low SS values (i.e. increased soil moisture) are observed for all GPR survey lines, suggesting fast lateral flow that traveled 2.2 m downslope in 30 min after the start of water release (TL4–TL1 in Figs. S1–S3). For the soil beneath GPR-Line 1, the magnitude of soil moisture buildup (indicated by the decline in SS values) came up to the maximum at 40 min after the last water release (Fig. 4; also see TL8–TL1 in Fig. S1) and began to decrease as the hydraulic potential gradient, the driver of water flow through the soil, reduced. The peak of soil moisture buildup due to stemflow input was delayed to 75 min and 105 min after the last water release in the soil below GPR-Line 2 (Fig. 4; also see TL9–TL1 in Fig. S2) and GPR-Line 3 (Fig. 4; also see TL10–TL1 in Fig. S3), respectively. The delayed response demonstrates the development of

downslope transport of stemflow water under matric potential gradient (Fig. 4).

3.2. Time-lapse ERT

Time-lapse ERT images are also compared before water release, immediately after the last water release, and under the final condition in Fig. 5. A complete comparison of time-lapse ERT images is provided in Fig. S4. Under the initial condition, ERT-Line 1 contains an area of low resistivity from 1.7 to 3.2 m along the survey line (Fig. 5a), likely corresponding to the root zone that is wetter than surrounding soil matrix. Time-lapse images of ERT-Line 1 reflects non-uniform stemflow transport that resistivity decreases from horizontal locations of 0.5 to 3.0 m (Figs. 5b and S4a–f). Right after the first pulse of water release, ERT-Line 1 shows negative changes in resistivity from 2.0 to 2.5 m along the survey line (Fig. S4b), indicating fast water transport from the base of the tree to ERT-Line 1. This localized wetting zone (i.e. preferential flow influencing zone) exists throughout the experiment period (Fig. S4a–f). Following the third pulse of infiltrated stemflow, another localized wetting zone appears from 0.7 to 1.2 m along the survey line (Fig. S4b) and persists until the end of the experiment (Fig.

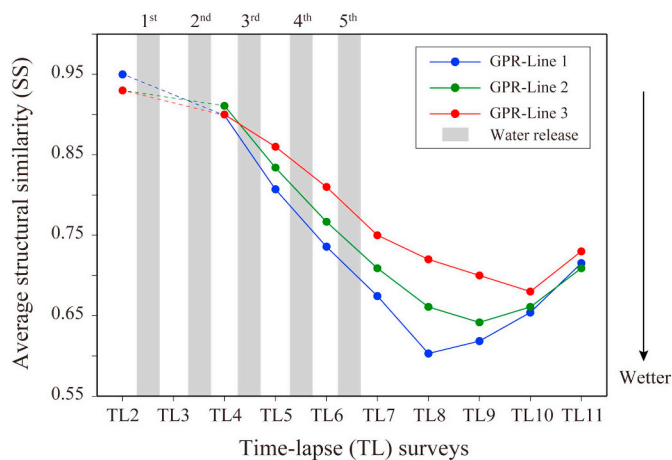


Fig. 4. Average similarity attributes (SS) of time-lapse (TL) GPR images calculated between the pre-wetting image and that collected during each of the time-lapse surveys for all GPR survey lines. Lower SS values indicate the higher deviation from the pre-wetting condition, which is likely caused by the increase in soil moisture due to subsurface preferential flow. Note that the GPR data collected immediately after the first water release (TL3) is problematic and is excluded for analysis.

S4g). Under the final condition, time-lapse images show slightly less extensive zones of decreased resistivity below ERT-line 1 (Fig. 5e) than right after the completion of the last water release (Fig. 5d). Areas with decreased resistivity in ERT images appear to spread laterally at a depth of ~ 0.7 m (Fig. 5b, d). The less permeable C horizon and clay accumulation likely restricted deep percolation of stemflow water and thereby triggered lateral flow (Schwärzel et al., 2012; Spencer and van Meerveld, 2016).

The overall extent of soil moisture buildup beneath ERT-Line 2 is much more limited than below ERT-Line 1 (Figs. 5f–j and S4h–n), implying that only limited stemflow water arrived at ERT-Line 2 (2.8 m downslope from the study tree) at the depth extent investigated. Time-lapse ERT images reveal three major localized wetting zones in the soil, from 0.3 to 1.3 m, 2.2 to 2.7 m, and 3.7 to 4.7 m in the horizontal distance along the survey line (Fig. 5h, j). Different from the observation from ERT-Line 1, more extensive zones of decreased resistivity appear under the final condition below ERT-Line 2 (Fig. 5j) than right after the last water release (Fig. 5h). This indicates the development of downslope transport of stemflow water under matric potential gradient without additional water input. The lateral extension of the area with negative changes in resistivity at the boundary to the C horizon (~ 1 m in depth in Fig. 5h, j) indicates the generation of lateral flow at the surface of a less permeable layer, which is consistent with the results reported in Tromp-van Meerveld and McDonnell (2006) and Spencer and van Meerveld (2016).

3.3. Comparison between GPR, ERT, and ground truthing

The initially wetted area identified beneath ERT-Line 1 (Fig. 5a) matches the clusters of lateral roots detected in upslope GPR survey line (Fig. 3d), reflecting the spatial concentration of soil water to the root zone due to hydraulic redistribution via roots. As water releases continued, resistivity values in the soil decreased from the center of the survey line at 3.0 m and gradually extending to 0.5 m along the survey line (Figs. 5a–e and S4a–g), which agrees with the wetting patterns revealed in GPR-Line 1 (Fig. 3e–f). The two localized stemflow routing zones revealed in ERT images (centered at ~ 0.9 m and ~ 2.2 m along the survey line; Figs. 5a–e and S4a–g) are consistent with the two wetting areas identified below GPR-Line 2 (the blue dashed box in Fig. 3g–i). The initial condition of ERT-Line 2 suggests a more homogeneous pattern of soil moisture (Fig. 5f), which is in line with the more

uniform distribution of lateral roots identified by GPR imaging (Fig. 3p). Following the initial infiltration of stemflow, localized subsurface routing appears 30 min after the start of the first water release, i.e. a flow velocity of 5.6 m/h (Fig. S4j), confirming the formation of the fast subsurface lateral flow identified in GPR images (Figs. 4 and S3).

After the last water release during the experiment, the arrival of stemflow water along a coarse root at 50 cm depth was observed in the soil trench (Fig. 6f) at around 1 m downslope of ERT-Line 2. This is in line with the wetting area identified in the lower right portion below ERT-Line 2 (Fig. 5h–j). Exposed near-surface lateral roots confirm the distribution pattern of roots revealed in the GPR image collected over GPR-Line 2 (Fig. 6a, b). Comparison between observation in all excavated soil pits and the corresponding GPR images indicates that only coarse roots (> 1 cm in diameter) could form recognizable hyperbolic reflections under the field condition with the GPR system used. For example, one major preferential flow path identified in GPR images (Fig. 3k, l) and ERT images (Fig. S4c–j) was verified as a result of a coarse lateral root (3.5 cm in diameter) and the macropore beneath it that were observed in the excavation (Fig. 6c–e). In the soil trench, we also observed the roots of a surrounding tree growing into the geophysical survey area, especially under GPR-Line 1 (Fig. 6f). However, these roots were growing perpendicular to the slope direction, i.e. parallel to GPR survey direction. Previous studies suggested that GPR is unable to detect lateral roots if the survey direction is parallel to the rooting direction (Guo et al., 2015). In the geophysical survey area downslope of GPR-Line 1, there were no observable roots from nearby trees (Fig. 6f). We did not observe other buried objects that could form hyperbolic reflections in GPR images, such as boulders. Observation in the excavation confirmed the effectiveness of the proposed method to identify and locate coarse roots of the study tree.

4. Discussion

4.1. The role of the belowground funneling on soil moisture and hillslope hydrology

Even though stemflow may be a small percentage of incident rainfall, it almost always constitutes a larger input per unit area than either rainfall or throughfall (Carlyle-Moses et al., 2018). In fact, stemflow funneling ratios often exceed unity even when stemflow is $< 1\%$ of incident rainfall (Carlyle-Moses et al., 2018). Our results showed that after water infiltrates into the soil, some of the stemflow can further be funneled along laterally-oriented coarse roots, forming fast, downslope lateral flow (Figs. 3 and 5). While Johnson and Lehmann (2006) suggested that stemflow concentrates precipitation at the base of the tree where preferential transport of stemflow water via soil macropores around tap roots takes place, which enhances the deeper and faster percolation of precipitation to the deeper soil (e.g. Li et al., 2009; Schwärzel et al., 2012) and even the groundwater (Tanaka et al., 1996), our results signify a variant of double-funneling whereby lateral roots are also important funnelers of stemflow in the hillslope soil (Figs. 3 and 5).

Lateral flow of stemflow along roots can cause a higher increase in soil moisture at depth than that at the soil surface and replenish soil moisture further away from the near-stem soil. This effect changes the partition of the stemflow input between vertical percolation and lateral transport in the subsurface (Spencer and van Meerveld, 2016), which affects the spatial distribution of soil moisture (Li et al., 2009; Liang et al., 2011) and can explain the different soil microbial communities between high and low stemflow producing trees species (Rosier et al., 2016). Previous stream chemistry studies at the same catchment recognized soil water as a component of storm-event runoff in the stream (Inamdar et al., 2013). The stemflow routing experiment conducted in this study demonstrated the high infiltration rate of stemflow water into the soil that recharged the soil water. Lateral root-induced preferential flow can be a possible way to direct stormwater into the downslope

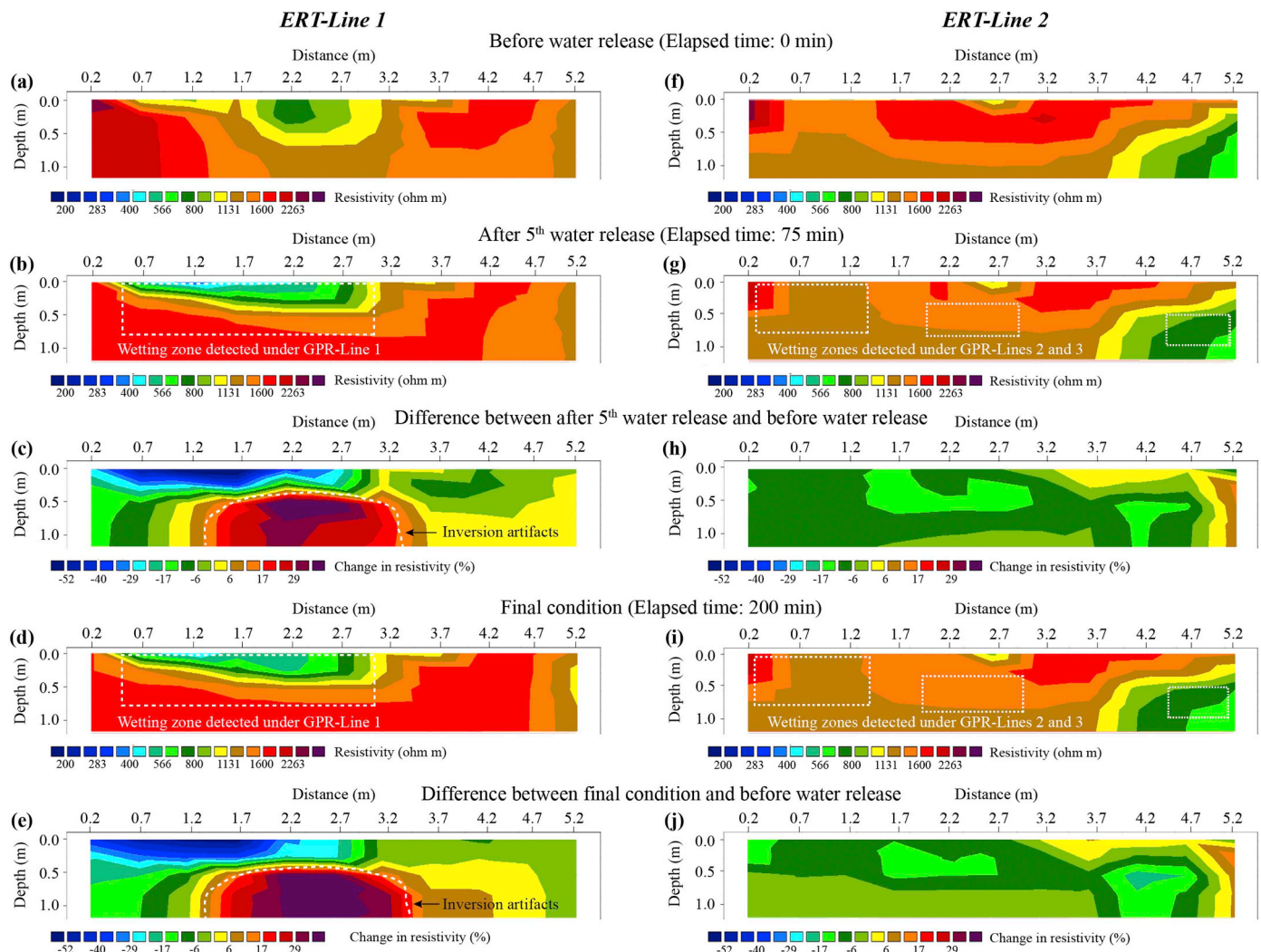


Fig. 5. Time-lapse ERT images collected (1) before stemflow application over ERT-Line 1 (a) and ERT-Line 2 (f), (2) immediately after the fifth water release (b and g), and (3) under the final condition (d and i). (c and h) show the percent change in resistivity between the final condition and the pre-wetting condition. (e and j) show the percent change in resistivity between the final condition and the pre-wetting condition. Dashed rectangles in b, d, g, and i indicate the wetting zones identified in corresponding upslope GPR images (i.e. the blue dashed boxes in Fig. 3; GPR-Line 1 for ERT-Line 1 and GPR-Lines 2 and 3 for ERT-Line 2, respectively). A negative percent change from pre-wetting condition (indicated by green-blue areas) represents actual stemflow routing, whereas a positive percent change forming right below the wetting zone (indicated by the yellow-red areas below the dashed lines) is due to inversion artifacts. The appearance of resistive anomaly artifacts below conductive anomalies is likely caused by the preferential flow through the now less resistive near-surface. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

stream (Schwärzel et al., 2012). Some prior work suggested the sub-surface lateral transport of stemflow along roots can be a likely mechanism in the runoff generation in steep hillslopes, such as on a forested hillslope with a slope gradient of greater than 28° (Liang et al., 2011). Our results imply that even on gently-rolling hillslopes, such as the hillslope investigated here with a slope gradient of 12° , below-ground funneling of stemflow along lateral roots have the potential to contribute to stormwater runoff and stream generation. This finding supplements our understanding of the interplay between roots and soil water. On the one hand, root water uptake transports water from deep soil layers to the root zone (i.e. hydraulic lift). On the other hand, soil macropores around coarse roots can act as preferential pathways that direct water bypassing the root zone soil and recharges the deeper soil or contributes to stream runoff.

4.2. Pairing GPR and ERT to characterize the belowground funneling

Geophysical techniques have emerged as valuable means for minimally invasive determination of the subsurface at a variety of settings

and scales (Jayawickreme et al., 2014; Parsekian et al., 2015). Time-lapse geophysical approaches, in particular, provide a new suite of tools to monitor the dynamic interplay between plants and soils (e.g. Boaga et al., 2013). The advantage of time-lapse geophysical methods for studying plant-soil interactions over traditional means includes (1) revealing both the spatial distribution and temporal dynamics of sub-surface processes, e.g. flow along coarse roots, by repeated surveys instead of one-time result obtained by invasive methods, e.g. dye experiments and (2) covering a range of spatial and temporal scales from the pedon to the hillslope and catchment scales instead of point-based measurements, e.g. soil coring. Time-lapse GPR used in this study also improves the accuracy of identifying coarse roots by selecting the hyperbolic reflections that constantly appear in successive surveys to eliminate the influence of random noise signals (Fig. 3). Under field conditions, it is challenging to suppress all of the noise signals in GPR images (e.g. bad antenna/ground coupling, system noise, or interference from nearby radio transmitters), which influence the shape of a root's signal and thus the detection and quantification of roots (Li et al., 2016). Because noise signals are random and likely do not repeat in

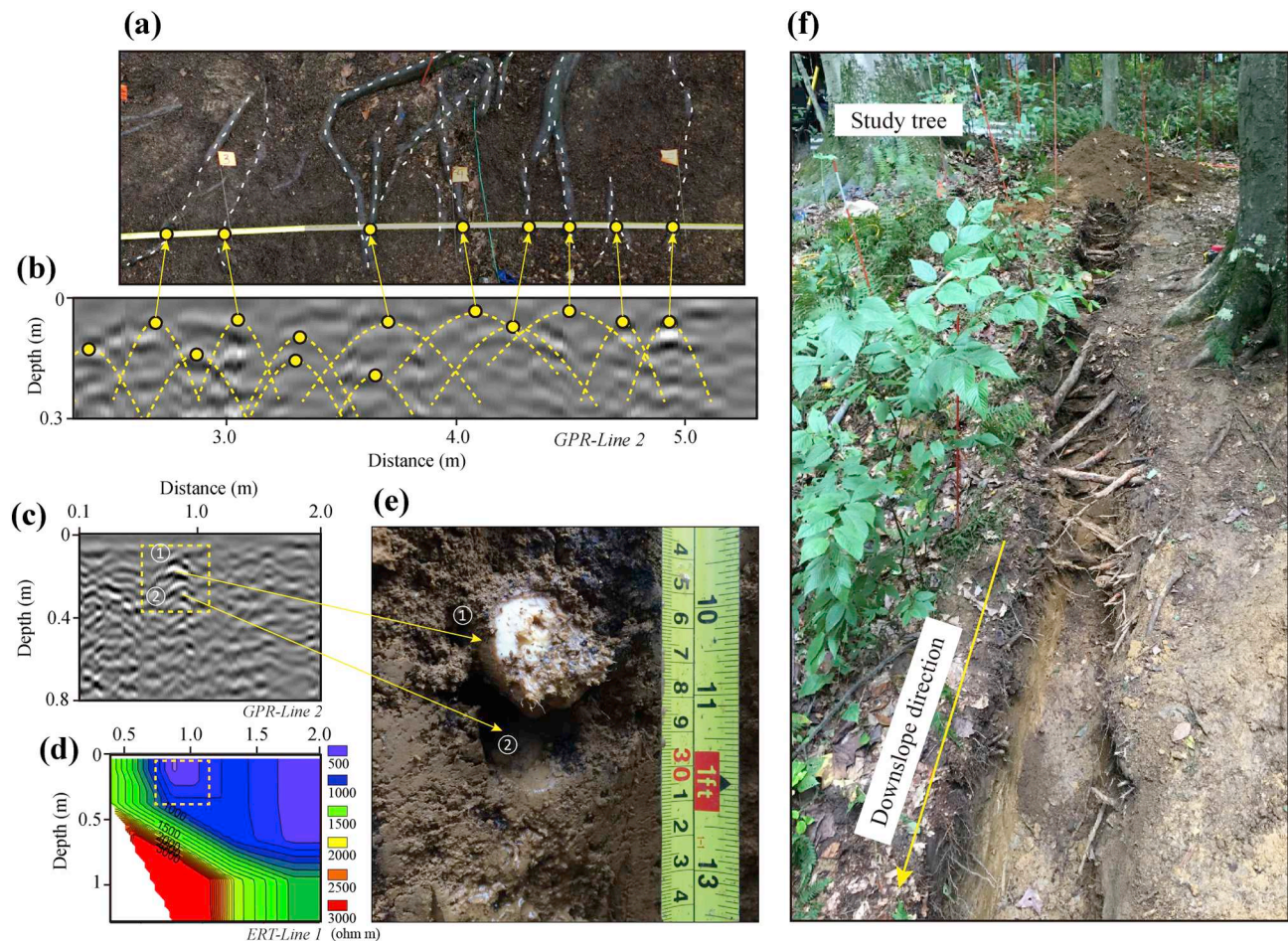


Fig. 6. Ground truthing of the geophysical results. (a) A ground view of the upslope section of the geophysical survey area was excavated to expose and document lateral roots distribution, indicated by the white dashed curves. The yellow tape indicates the location of GPR Line 2. The yellow flags indicate 3, 4, and 5 m along the survey line. (b) The GPR image (a section view of the soil profile) collected over GPR-Line 2 immediately after the fifth water release. The arrows indicate the correspondence between the lateral roots detected in the GPR image and that revealed in the excavation (a). (c–e) A lateral root-derived preferential flow path revealed in geophysical images (indicated by the yellow dashed boxes in c and d; also see Figs. 2k, l and 3e–g) that is groundtruthed by excavation. The hyperbolic reflections in the GPR image (c) are produced by a lateral root (①) and the macropore below the root (②). The ERT image also reveals a wetting area at the same location. (f) A soil trench was opened along the slope, showing the lateral spread of coarse beech roots from the surface to 40 cm in depth, which is in line with the pattern identified in GPR images (Fig. 3d, j, p). About one hour after the last water release, we observed the arrival of stemflow water along a lateral root on the wall of the soil trench at 50 cm depth, which was 1 m further downslope of ERT-Line 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

GPR images in successive surveys, time-lapse GPR allows one to separate root signals (constantly occurring in repeated GPR images) from random noise and choose appropriate root signals (symmetric and intact hyperbolae) to locate roots.

However, one limitation of nearly all geophysical measurements (including GPR and ERT adopted in this study) is the uncertainty associated with data interpretations due to the indirect nature of the measurement. For example, GPR measures the contrast in dielectric permittivity in the subsurface. Although moisture change is the key control of subsurface dielectric permittivity (Parsekian et al., 2015), the geometric mismatch between time-lapse GPR images also contributes to signal changes, which influence the repetitiveness of signal changes in time-lapse GPR images to indicate soil moisture dynamics (Guo et al., 2014). Whereas signal changes in time-lapse ERT are mainly caused by moisture dynamics (Michot et al., 2003). Pairing GPR and ERT reveals subsurface areas showing moisture change and the temporal trends of moisture dynamics (by ERT) and pinpoints flow pathways and their relative position with lateral coarse roots (by GPR).

Moreover, pairing GPR and ERT helps explore the relationship between roots and soil more than using a single geophysical technique. For example, a cluster of roots is identified beneath GPR-Line 1

(Fig. 3a). The ERT survey also suggests this area was initially wet before any stemflow inputs (Fig. 5a). The agreement between GPR and ERT confirms the extent of rhizosphere and the root distribution pattern. Without the supplementary data from ERT, GPR image cannot reveal the moisture gradient between rhizosphere and surrounding soil matrix. Without root distribution derived from GPR, ERT image cannot link the wetter area to the presence of roots. Therefore, combining GPR and ERT enhances more reliable interpretations and ensures the non-invasive manner of geophysical investigation of the belowground funneling and other subsurface properties and processes (Leucci and De Giorgi, 2005).

Previous dye staining experiments have visualized preferential flow along heart and sinker roots (Li et al., 2009), laterally oriented coarse roots (Liang et al., 2011), and decayed root channels in the soil (Schwärzel et al., 2012; Spencer and van Meerveld, 2016). Previous GPR studies also suggested that the subsurface flow network was promoted by the development of lateral roots (Guo et al., 2014). Results of this study confirmed lateral roots as important pathways to transport stemflow water in the vadose zone. Different from the dye staining experiments that only show the one-time results after the event (Allaire et al., 2009), the experimental protocol reported here allows repeated monitoring of the routing of stemflow after different rainfall events to

identify the lateral roots that contribute to subsurface stemflow transport from those that do not (e.g. Fig. 3).

Mapping the routes of stemflow inputs in the subsurface is important for ecohydrological and hydropedological studies, which attempt to better understand the patterns and mechanisms of non-uniform routing in the soil as well as the partitioning of precipitation among infiltration and subsequent preferential routing in the subsurface, soil moisture recharge, and overland flow. Current hydrological models find it difficult to simulate soil moisture patterns after precipitation due to their lack of consideration of actual infiltration areas and subsurface flow pathways (Cheng et al., 2018; Guo and Lin, 2018). Therefore, the proposed method can advance the characterization of localized stemflow inputs and their preferential transport in field soils to gain new insights into the complexity in subsurface water flow and to develop a robust and realistic hydrologic conceptualization to guide improved hydrologic modeling and forecasting (Beven and Germann, 2013; Lin, 2010).

4.3. Further improvement of the proposed methodology

Our stemflow experiment involved applying water in five pulses, which allowed all of the stemflow to infiltrate at the base of the study tree and avoided overland flow. The presence of surface water can impair the performance of GPR and ERT in probing subsurface moisture dynamics. The stemflow was equivalent to a rainfall event with an intensity of 12 mm per hour. The high intensity favored stemflow infiltration (Levia et al., 2010) and the subsequent generation of preferential flow (Guo and Lin, 2018), which was useful for identifying the areas of funneling in the soil. Our intermittent application of stemflow mimicked the episodic and dynamic flows characteristic of American beech under variable rainfall intensities (Levia et al., 2010) and ensured the development of belowground funneling that was detectable by time-lapse geophysical surveys. Nevertheless, a more realistic representation of stemflow under natural conditions should be considered in follow-up studies. Because operating GPR during the rainfall is risky, and the detection depth decreases considerably with existing surface water. Pairing GPR and ERT to monitor the belowground funneling is challenging when it rains. We suggest collecting ERT surveys during the rainfall to monitor the initial response of soil moisture to rainfall and both GPR and ERT data after the event to link areas with moisture changes with root distribution to reveal potential pathways for belowground funneling.

In addition to a short period right after the rainfall, we also suggest to repeat the geophysical surveys for a period of several days after the event to monitor a complete process of stemflow infiltration and redistribution. Although previous studies have explored time-lapse GPR for monitoring subsurface flow, not much investigation of optimal timing for collecting GPR data after a rainfall event has been conducted. This is a difficult problem because evolving soil moisture patterns that drive GPR signal changes also depend on the rainfall characteristics and antecedent moisture conditions (Guo and Lin, 2018). The recent development of a long-term, low-cost, and real-time ERT system (Chambers et al., 2016) allows continuous ERT data collection for months at a time to monitor electrical resistivity dynamics associated with the full cycles of wetting and drying over many rainfall events. The ERT images would be processed and reviewed to identify the optimal times, e.g. with the largest magnitude of moisture changes, to most effectively collect GPR surveys to determine the belowground funneling. In addition, the spacing between electrodes can be shortened to maximize the vertical resolution of ERT to help identify localized preferential flow pathways. However, a denser distribution of electrodes often requires more time to complete each time-lapse survey and impairs the time resolution of monitoring subsurface flow dynamics.

Under ideal conditions for GPR performance, e.g. dry sandy soils, GPR shows potential to estimate the diameter of an individual coarse root (Guo et al., 2013; Hirano et al., 2009). Linking root size to the

frequency of preferential routing along roots can help understand the mechanism of the belowground funneling. However, for this study site and the GPR system used, the proposed geophysical method is difficult to detect roots (< 1 cm in diameter) and investigate their role in conducting subsurface preferential flow.

Improving the processing protocol of time-lapse geophysical data, such as the structural similarity attribute used here (Angermann et al., 2017; Jackisch et al., 2017), also strengthens the efficiency of geophysical methods in revealing the ecohydrological dynamics in the soil. Establishing the quantitative relationship between structural similarity (or electrical resistivity) and soil moisture based on site-specific conditions can further enhance time-lapse geophysical survey in studying the belowground funneling (Michot et al., 2003). In this study, an increase in resistivity in a layer right below where the resistivity decreases forms after the infiltration of the stemflow (Fig. 5c, e), which is a common artifact of time-lapse inversion of resistivity data (Clément et al., 2009). Because reliable resistivity inversion is difficult below a conductive layer (the localized wetting zones due to preferential transport of stemflow in this case), and often the inversion algorithm makes the conductive layer too conductive and produces a resistivity increase below it even though the increase in resistivity beneath the conductive layer is not physically present (Carey et al., 2017). Recent developments in resistivity inversion methods (e.g. Carey et al., 2017) can suppress the inversion artifacts and enhance the use of time-lapse resistivity to obtain reliable estimates of subsurface flow processes.

We believe that further enhancements of geophysical methods, especially the integration of multiple techniques and post-processing of geophysical data, can help in-depth understanding belowground interactions between biomass, pedosphere, and hydrosphere with an improved time-space resolution. Given the increasing use of geophysical techniques for ecohydrological studies (Jayawickreme et al., 2014; Parsekian et al., 2015), we expect broader applications of our experimental protocol (i.e. the paired geophysical imaging techniques) to facilitate the enhanced understanding of the complex plant-soil interactions and the continuity of the ecohydrological process between the aboveground and the belowground parts of the Critical Zone under different conditions and across various environments.

5. Conclusions

The double funneling model signifies the important role of trees in the translation of precipitation to flow in the soil, both aboveground and belowground. The refined experimental setup of pairing GPR and ERT reported here captures the natural flow processes along the trunk and roots and provides a field method to visualize plant-soil interaction on subsurface water movement. Combining multiple geophysical techniques enhanced the reliable and non-invasive investigation of belowground funneling. Our interpretations of geophysical images have been further ground-truthed by in situ observations and follow-up field excavations. Despite the gentle slope gradient, fast downslope transport of stemflow input along lateral roots was visualized and confirmed. To the knowledge of the authors, this is the first attempt to combine GPR and ERT in tandem to examine the temporal dynamics and spatial distribution of stemflow in the subsurface. These findings enhance the understanding of the significant role of trees in water flux through the Critical Zone, such as the funneling effect of tree roots on belowground water redistribution.

Declaration of competing interest

The paper has no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoderma.2019.113953>.

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